

PPS 11.1 Waves

1) C, upwards/downwards



2) Wave moves $\approx 0.8 \text{ m}$ in 0.2 s

so moves 4 ms^{-1}

wave length $\approx 0.2 \text{ m}$

$$f = \frac{v}{\lambda} = \frac{4}{0.2} = 0.64 \text{ Hz}$$

③

3) $2 \text{ ms} = 2 \times 10^{-3} \text{ s}$

$$340 \times 2 \times 10^{-3} = 0.68 \text{ m}$$

4) Point C.

5) $f = \frac{1}{T} = 0.25 \text{ Hz}$

$$v = 5$$

$$v = f\lambda$$

$$\lambda = \frac{v}{f} = 5 \times 4 = 20 \text{ m}$$

6) $1200 \text{ km} = 1.2 \times 10^6 \text{ m}$

$$v = 5 \text{ ms}^{-1}$$

$$t = \frac{d}{v} = \frac{1.2 \times 10^6}{5} = 240,000 \text{ s}$$

$$\frac{240,000}{3600} = 67 \text{ hours}$$

6) a)

i) $y_0 = 0.2 \text{ m}$

ii) $\lambda = 1.4 \text{ m}$

b)

i) in 0.25 s , wave moves $\approx 0.2 \text{ m}$

$$\therefore c = \frac{0.2}{0.25} = 0.8 \text{ ms}^{-1}$$

ii) $v = f\lambda$

$$f = \frac{v}{\lambda} = \frac{0.8}{1.4} = 0.57 \text{ Hz}$$

$$T = \frac{1}{f} = 1.75 \text{ s}$$

7)

$$\lambda = 30 \text{ cm}$$

$$T = 2 \times 2.5 \text{ ms} = 5 \text{ ms} = 5 \times 10^{-3} \text{ s}$$

$$f = \frac{1}{T} = 0.2 \times 10^3 = 200 \text{ Hz}$$

8)

$$n_i \sin \theta_i = n_r \sin \theta_r$$

$$1 \times \sin 40^\circ = n_r \sin 60^\circ$$

$$n_r = 1.35$$

$$c = \frac{1}{1.35} \times 3 \times 10^8 = 2.2 \times 10^8 \text{ (B)}$$

9)



$$\frac{1}{n_r} = \frac{\sin \theta_r}{\sin \theta_i} = \frac{\sin 20^\circ}{\sin \theta_r}$$

$$\theta_r = 26^\circ$$

10)

a) i) $\sin \theta_c = \frac{n_r}{n_i}$

$$= \frac{1}{1.309}$$

$$\theta_c = 49.8^\circ$$

ii. $\theta_c = \arcsin\left(\frac{1}{1.34}\right)$

$$= 47.3^\circ$$

iii. $\theta_c = \arcsin\left(\frac{1}{1.54}\right)$

$$= 40.4^\circ$$

11)

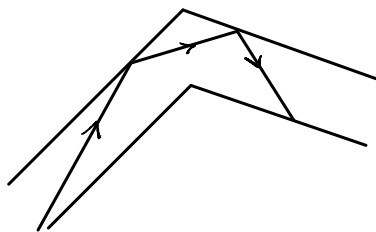
$$n_i = \frac{c}{2.2 \times 10^8} = 1.33$$

$$n_r = 1.52$$

$$\sin \theta_c = \frac{n_i}{n_r} = \frac{1.33}{1.52}$$

$$\theta_c = 61.1^\circ \text{ (D)}$$

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$$n = \frac{1.402}{1.074} = 1.014$$

$$\sin \theta_c = \frac{1}{1.014}$$

$$\theta_c = \arcsin(1.014)$$

$$= \underline{80.2^\circ}$$

13 $n_i = 98^\circ$

$$a) n_i \sin \theta_i = n_r \sin \theta_r$$

$$1 \times \sin 98 = 1.26 \sin \theta_r$$

$$\theta_r = \arcsin\left(\frac{\sin 98}{1.26}\right)$$

$$= 41.5^\circ$$

$$b) 1.28 \sin 41.5 = 1.34 \sin \theta_r$$

$$\theta_r = \arcsin\left(\frac{1.28 \sin 41.5}{1.34}\right)$$

$$= 39.3^\circ$$

14 $n_1 < n_3 < n_2$ A is false: $\theta_1 \neq \theta_3 \therefore c_1 \neq c_3$

$$B \text{ is true: } n_1 \sin \theta_1 = n_3 \sin \theta_3$$

$$\frac{c}{v_1} \sin \theta_1 = \frac{c}{v_3} \sin \theta_3$$

$$v_3 \sin \theta_1 = v_1 \sin \theta_3 \quad + \text{ o.t.c.}$$

C is false: f only changes when source oscillation changes

$$D \text{ is false: } \frac{n_2}{n_1} = \frac{\frac{c}{v_2}}{\frac{c}{v_1}} = \frac{v_1}{v_2}$$

$$= \frac{v_1}{v_2}$$

15) i) $v = f\lambda$

$$3 \times 10^8 = 4.64 \times 10^{-7} f$$

$$f = \frac{3 \times 10^8}{4.64 \times 10^{-7}}$$

$$= 6347 \times 10^{14} \text{ Hz}$$

$$1.53 = \frac{c_{\text{vacuum}}}{c_{\text{glass}}}$$

$$c = \frac{3 \times 10^8}{1.53}$$

$$= 1.96 \times 10^8 \text{ ms}^{-1}$$

$$ii) \lambda = \frac{1.96 \times 10^8}{6.347 \times 10^{14}} = 3.06 \times 10^{-7} \text{ m}$$

6

$$i) n_i \sin \theta_i = n_r \sin \theta_r$$

$$\frac{1.53}{2} = 1 \times \sin \theta$$

$$\theta = \arcsin\left(\frac{1.53}{2}\right) \quad (\theta = 30^\circ)$$

$$= 49.9$$

$$ii) \text{critical angle} = \arcsin \frac{1}{1.53} = 40^\circ$$

When $i > 40^\circ$, TIR occurs since $i > \text{critical angle}$.P will become $i = 60^\circ$.

Q will not exist, as all light will be reflected.